
LECTURE NOTES

Lecture 8: Beyond Local Optima

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1 Benchmarking

2 Stochastic Methods

- Randomness can help escape local minima
- Control over randomness and the exploration vs exploitation trade off.

2.1 Noisy descent

Saddle points, where the gradient is very close to zero, can cause descent methods to select step sizes that are too small to be useful

We add Gaussian noise at each descent step

$$x_{k+1} = x_k + \alpha \nabla f(x_k) + \varepsilon_k, \quad \varepsilon_k \sim \mathcal{N}(0, \sigma_k^2)$$

- $\sigma_k = \frac{1}{k}$

2.2 Stochastic Gradient Descent

We can approximately evaluate gradients using randomly chosen subsets of the training data (batches)

This is much less computationally expensive than calculating the true gradient at every iteration. It yields approximately the same effect as noisy gradient approximation.

This ensures that the positive step sizes be chosen such that:

$$\sum_{k=1}^{\infty} a_k = \infty, \quad \sum_{k=1}^{\infty} a_k^2 < \infty$$

Allows the step sizes to decrease and allow the method to converge, but not too quickly so as to become stuck away from a local minimum

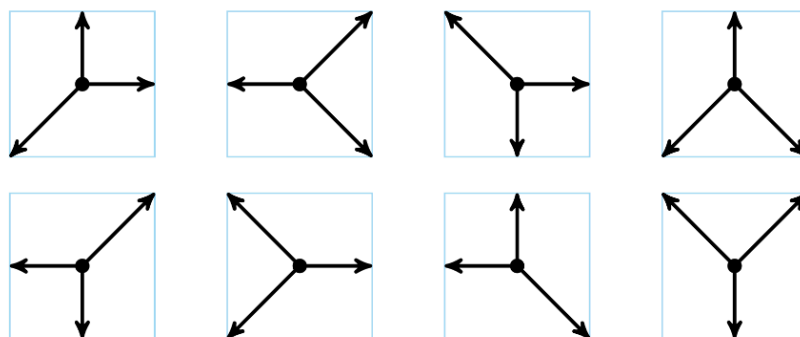


Figure 1:

2.3 Mesh Adaptive Direct Search

Similar to generalized pattern search but uses random positive spanning directions

Example: set of positive spanning sets constructed from nonzero directions $d_1, d_2 \in \{-1, 0, 1\}$.

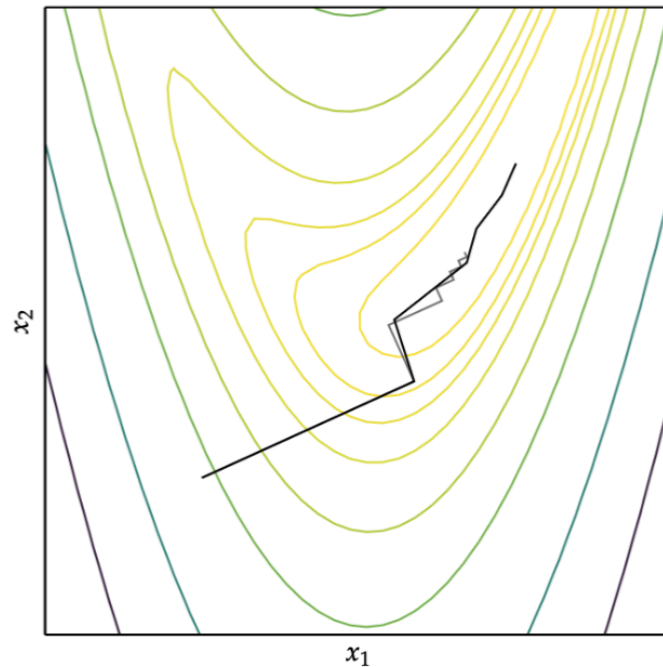


Figure 2:

2.4 Simulated Annealing

often used on functions with many local minima due to its ability to escape local minima.

A candidate transition from x to x' is sampled from a transition distribution T , eg, multi-variate Gaussian

$$x' = x + \varepsilon, \quad \varepsilon \sim T$$

- **Metropolis acceptance criterion:**

$$p(x, x') = \begin{cases} 1 & \delta \leq 0 \\ e^{-\frac{\delta}{t_k}} & \delta > 0 \end{cases}$$

2.5 Simulated annealing

Where $\delta = f(x') - f(x)$ is the change in the change in the objective value and t_k is the **temperature** at iteration k .

We can also do variable step-size v (separate direction components) in order to not use covariance matrix as a hyperparameter.

A cycle of random moves, one in each direction

$$x' = x + rv_i e_i$$

where r is randomly sampled from $\{-1, 1\}$

Then after n_s cycles, adjust the step size

$$\begin{cases} v_i \left(1 + c_i \frac{\frac{a_i}{n_s} - 0.6}{0.4} \right) & a_i > 0.6n_s \\ v_i \left(1 + c_i \frac{0.4 - \frac{a_i}{n_s}}{0.4} \right)^{-1} & a_i < 0.4n_s \\ v_i & \text{otherwise} \end{cases}$$

2.5.1 Annealing plan

t_k can decrease in different ways

- logarithmic annealing schedule

$$t_{k+1} = t_0 \cdot \frac{\ln(n)}{\ln(k+1)}$$

- exponential annealing schedule

$$t_{k+1} = \gamma t_k$$

- fast annealing

$$t_{k+1} = \frac{t_0}{k}$$

We can apply the decrease at different times.

Either at every proposition (either accepted or rejected), every move or at every m_t iteration.

2.6 Cross-Entropy Method

Cross-entropy is a measure of divergence between two probability distributions p and q . We measure cross-entropy in a case where one distribution (the one of optimal solutions) is unknown.

A model is created and then its cross-entropy is measured on the elite set to assess how accurate the model is in predicting this set.

2.7 Multivariate normal distribution

2.8 Solution of Max Likelihood

2.9 Natural Evolution Strategies

Similar to cross-entropy method, except instead of parameterizing distribution based on elite samples, it is optimized using gradient descent.

We know from statistics (Introduction to Machine Learning /w Melih) that

$$\mathbb{E}[f(X)] = \int_{\mathbb{R}} x \cdot p(x)$$

then

$$\mathbb{E}[f(x)] = \int_{\mathbb{R}} f(x) \cdot p(x|\theta)$$

2.10 Covariance Matrix Adaptation Evolutionary Strategy (CMA-ES)

Uses the same approach as natural evolution strategy and cross entropy method, but the proposal distribution is a multivariate Gaussian parameterized by a covariance matrix.

At every iteration, m designs are sampled from the multivariate Gaussian:

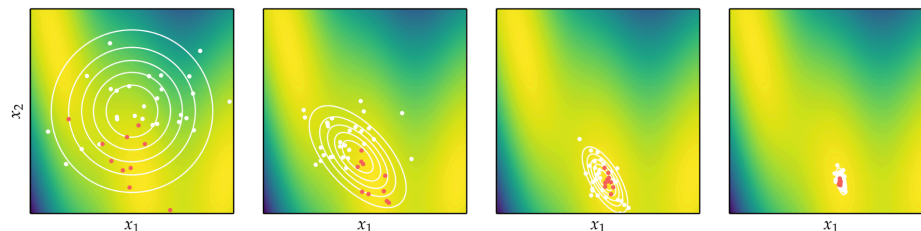
$$x \sim \mathcal{N}(\mu, \sigma^2 \Sigma)$$

Mean vector μ and a covariance matrix Σ with a step-size scalar σ

The scalar ensures that the covariance only increases or decreases in a single direction with every iteration.

2.11 Cross Entropy Method

Cross-entropy is a measure of divergence between two probability distributions p and q



By iteratively choosing the best candidates from a distribution, and updating that distribution a new distribution to the elite set using MLE and repeating until some convergence, the distribution moves closer and closer to the optimum.

- Limitations

- Only considers top fraction $\rightarrow$ ignores potentially useful information from other samples
- Can make abrupt jumps, which may be unstable or slow to adapt

2.12 Natural Evolution Strategies

Similar to CEM but this time we divergence

$$J(\theta) = \mathbb{E}_{x \sim [f(x)]}$$